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Abstract

Drinking-water monitoring programmes must decide what to measure next, where and with which instrument, often from archives, models and instrument documentation produced for other purposes. We developed an evidence-aware decision framework that links record harmonisation, validation design, external transfer and documentary appraisal of instruments, and returns for each monitoring context an action together with the evidence required to reassess it. We applied it retrospectively to 26 monitoring contexts in India, Kwale County (Kenya) and Malawi, which received eight distinct actions, including repeat measurement, data repair and paired chemistry, and to published evidence from Bangladesh and Ghana, where it permits kit screening with confirmation. Across seven water-quality tasks in a multinational fitting archive, the area under the receiver operating characteristic curve (ROC-AUC) fell from 0.9396 when records were split at random to 0.7696 when contiguous spatial blocks were held out, with one split per design. Separately fitted models kept their risk ordering in 5 of 13 external archive $\times$ parameter pairs, including four in Indian groundwater, which supports sequencing confirmatory sampling within those archives; calibration slopes were below 1 on all 13 pairs, so no transferred score was used as a probability. In the Indian archives, further records could change the classification of at least 17 of 160 eligible state or Union Territory $\times$ parameter cells at an assumed action level of 0.10. A documentary audit of 23 products found 891 of 1,242 product–criterion cells unreported in any located document and no product documenting all 11 active criteria, so no product was named. The framework records, for each action, the rule and evidence it rests on; the Kenyan

and Malawian archives are local, not national, and whether the framework improves routine monitoring has not been tested prospectively.

Introduction

Drinking-water quality monitoring reaches fewer people than it is meant to protect. Over the Sustainable Development Goal period, the share of the global population covered by total estimates of safely managed drinking water rose from 35% to 72%, and such estimates became available for India for the first time only in 2025 [1]. Surveillance is also less frequent than required: in urban areas, 21 of 102 countries reported drinking-water surveillance at 95–100% of the required frequency [2]. Infrequent sampling can overestimate how safe water is [3]. A programme with a fixed testing budget therefore has to decide which parameters to measure, at which water points and with which instrument, on evidence that is thin and unevenly spread.

Public archives record where and when testing happened, not the risk across a population. In India, the Central Ground Water Board’s current monitoring design samples every quality station once every five years before the monsoon and at least a quarter of the stations before and after the monsoon every year [4]. An exceedance rate computed from records collected under any such design is a property of the sampling as much as of the water. Measuring where the answer is already settled spends the budget without changing an action, while not measuring where the evidence is ambiguous leaves an action resting on ambiguous evidence.

One response is to model where contamination is likely. Predictive modelling can relate the probability that a contaminant such as groundwater arsenic exceeds a guideline value to soil, climate and terrain predictors, producing maps meant to guide further testing rather than replace it [5]. National models for India have mapped groundwater fluoride from 12,600 measurements [6] and groundwater arsenic, with the hazard maps proposed to inform the prioritisation of testing [7].

How such models are evaluated decides what they can be used for, and practice differs. Environmental records are spatially autocorrelated, so a split that separates rows at random estimates something other than performance in an unseen region [8], and revisited sites compound the problem. Ignoring dependence between nearby samples biases model assessment towards optimism [9]; the national Indian models were evaluated on random training and test splits [6, 7]. Discrimination and calibration are likewise different properties: a model can order locations well while its stated probabilities are distorted [10]. Estimated risks can be unreliable even when discrimination is good [11], so a model whose ordering holds in a new setting may still misstate probabilities there.

Technology evidence raises the same question at the next step. Technology assessment characterises what instruments can measure and what deploying them demands, from power supply in remote settings to cleaning, signal drift and cost [12]. A programme needs performance near the threshold at which it must act. A 2003 verification of one arsenic field kit, in which one technical and one non-technical operator analysed

identical samples, found false-negative rates of 62% and 33% respectively when they read its colour charts, relative to 10 µg/L [13]. A microbial water-quality test has been costed at 21.0 ± 11.3 USD on average across 18 African monitoring institutions in six countries, once consumables, equipment, labour and logistics are counted, with the $\pm$ not defined in the source [14].

Decision analysis makes trade-offs between objectives explicit and can carry uncertainty through to the comparison of alternatives [15]; probabilistic forms report how probable each rank order is [16]. A ranking method returns a leader whether or not the criteria behind it are documented. Machine-learning models always make a prediction unless they are designed to abstain [17].

Monitoring decisions therefore depend on several bodies of evidence, each of which can be strong enough for one use but not the next. Part of this difficulty is recognised: a review of machine-learning prediction of United States drinking-water contaminants named accounting for spatial autocorrelation in model training among the remaining methodological challenges [18]. The nearest existing approaches are the fixed surveillance schedules and the model-guided prioritisation of testing cited above. We developed an evidence-aware decision framework that connects record harmonisation, validation design, external transfer and documentary appraisal of instruments, tests at each step whether the evidence supports the use a programme would make of it, and returns for each monitoring context an action together with the evidence required to reassess that action.

We applied the framework retrospectively to archived and published evidence: India as the principal application, local archives from Kwale County in Kenya and from Malawi, and decisions in Bangladesh and Ghana that rest on published evidence alone. The application addressed four linked questions: how much validation design changes the performance estimate a programme would rely on; whether a model's ordering and its probabilities carry over to groundwater archives that took no part in fitting; where further records could change a monitoring classification; and whether the documentation of named instruments is sufficient to compare them.

Materials and methods

Framework architecture and decision rules

The framework takes three bodies of evidence (an accepted monitoring archive, the output of any predictive model, and the documentary evidence about candidate instruments) and returns two things for each monitoring context, a country $\times$ source archive $\times$ parameter combination (Fig 1). The first is an action, which names what a programme should do next rather than scoring a place. Actions run from acquiring a

baseline or repairing unusable records, through prioritising confirmatory sampling, to direct or targeted
reference measurement and sentinel monitoring paired with chemistry; the Results give the action returned
for each context, and S2 Table defines all eight. The second return is the evidence required to reassess the
action: the additional evidence that could move the context to a different action. It is stated as a kind of
evidence, not as a sample size, cost or optimal design. Three gates stand between the inputs and the output,
and a context meets them in order: the validation gate at step 6 of the workflow, the permission gate at step
7 and the evidence gate at step 8.

Fig 1. The evidence-aware decision framework. Evidence strands are kept apart (step 1), records are
harmonised (2) and quantified (3), and documentation for 23 products is classified against 54 criteria (4).
Model output meets a validation gate (5 and 6) and a permission gate separating rank ordering from
probability (7); the evidence gate requires all 11 active criteria to be documented before a product is named
(8). Outputs are kinds of action, not the eight actions returned. MCDA, multi-criteria decision analysis;
CGWB, Central Ground Water Board.

The validation gate settles in which role a model may enter a decision: ordering locations so that
confirmatory sampling is sequenced, supplying a probability to compare with an action level, or neither. The
role follows from the validation design, not from the size of a reported metric. Monitoring records are not
exchangeable rows, because sites are revisited and nearby observations share geology, land cover and often a
sampling programme, so a random split mostly measures interpolation within the existing sampling structure.
Keeping every record from a site together removes the repeat-visit component; holding out contiguous spatial
blocks additionally requires the model to generalise across geography [8]. The gate prevents a random-row
figure from being read as performance in a district that contributed no training station.

Discrimination asks whether the model orders cases correctly; calibration asks whether a stated
probability matches the frequency actually observed at that probability [10]. Calibration here always means
model calibration, not the calibration of an instrument. A ranking tells a programme where to look first, and
a probability compared with a threshold tells it when to act. The permission gate therefore recognises three
states. Where ordering survives a move to a new setting but calibration there has not been shown to be
adequate, the model may set the order of confirmatory measurement and nothing more. Where ordering
survives and calibration on local labels from that setting has been assessed and found adequate, or restored
by local recalibration, a probability may also be compared with a fixed action level. Where ordering does not
survive, the model contributes nothing and direct measurement is required. No adequacy criterion for
calibration was fixed in advance in this study. The gate is applied to each archive $\times$ parameter pair, because
two archives holding the same analyte can return opposite verdicts.

Once a decision requires direct measurement it becomes a question about an instrument, and the evidence

gate governs it through two rules. The documentation threshold asks whether a criterion is documented for enough products for a comparison on it to be informative. The sufficiency rule asks whether a given product documents every active criterion and passes its hard analytical, matrix, deployment and identity requirements. A product that fails the sufficiency rule is not scored, and the framework returns a product-level abstention together with the documents that would lift it. Scoring an undocumented field at a middling value would reward weak disclosure, because a device that reports a poor maintenance interval would then rank below one that reports nothing. Missing documentation is unknown evidence about a product, not evidence that it performs poorly. Multi-criteria comparison of technology archetypes gives preference context for a class of instrument under stated weights, is not screened for the analytes a context needs, and establishes nothing about any product within that class.

We distinguish three monitoring functions, because comparing devices before the function is fixed ranks alternatives that were never doing the same job. Distributed spot screening covers existing water points. A higher-frequency sentinel at selected sites buys temporal resolution at the cost of power, logging, drift, fouling and downtime. Confirmatory or reference measurement establishes a contaminant-specific result. The functions combine within one monitoring architecture, no technology class is presumed to serve all three, and each setting's action names the function it relies on.

Abstention is accordingly a designed output with the same role as the reject option of a selective classifier [19], but triggered by missing evidence rather than by expected error (see Discussion), and not a failure. Because the gates are ordered, a context whose records cannot be accepted returns a baseline or repair action before any model is considered, one whose model output fails the validation or permission gate returns direct measurement, and one that clears both still reaches the evidence gate before any instrument is named. The rules are the same whatever a product documents, so no product's values can alter the gates it is judged by. How the gates were implemented for the contexts reported here, including which outcomes the authors recorded rather than computed, is set out below under Technology evidence and action rules.

Study design and unit of analysis

The unit of analysis for decisions is the monitoring context, one country $\times$ source archive $\times$ parameter combination, and 26 were analysed, 8 in India, 8 in Kenya and 10 in Malawi (S1 Text gives how they were generated). The Kenyan and Malawian archives are bounded local archives, not national samples, and no result from them is a national estimate for either country. Bangladesh and Ghana are applications of the decision logic to published evidence, with no model fitted or transferred; their studies were chosen for the decision each setting poses rather than found by a systematic search, so other studies could change the

evidence those applications rest on. Five populations are kept separate and never merged: 23 exact commercial products, a 33-entry technology landscape (Fig 1), reference laboratory methods, emerging research classes, and ten technology archetypes, classes of instrument compared by multi-criteria decision analysis (MCDA). Missing evidence was retained, not imputed.

Data sources and evidence roles

Six evidence strands enter the analysis (Fig 2): the fitting archive, the Indian archives, the United States archive, the Kenyan and Malawian archives, published studies from Bangladesh and Ghana, and documentation for named products. The fitting archive is the GEMStat global water-quality database of the UNEP GEMS/Water Programme [20]: a multinational archive, predominantly of surface water, used only for model fitting and internal evaluation. Seven classification tasks were built from it, for arsenic, nitrate, conductivity, suspended solids, turbidity, fluoride and pH (thresholds in S1 Text).

Fig 2. Evidence roles by country. Shading shows the role of each country’s evidence, not the extent of sampling: India holds the two Central Ground Water Board archives; the United States Water Quality Portal well-site extract, from four states, is the second external arm; Kenya and Malawi hold bounded local archives; Bangladesh and Ghana enter through published evidence only. GEMStat and Indian station locations are not drawn: their redistribution terms are unconfirmed. Boundaries: Natural Earth (public domain). Plotted values: S9 Data.

Two external arms were used to evaluate transfer, and neither took part in fitting. The primary arm is Indian groundwater from the Central Ground Water Board (CGWB), in two products that share a producer: National Water Data Portal records for pH, conductivity, total dissolved solids (TDS) and arsenic (the CGWB portal), and fluoride and nitrate tables recovered from the CGWB Ground Water Quality Data documents (the CGWB Year Book) [21–24]. CGWB samples groundwater from observation and monitoring wells, with the stated aim of assessing shallow aquifers against the Indian drinking-water standard [25], so a record describes groundwater at a monitoring well, not water as supplied. The second arm, WQP, is a 2023 well-site extract for four states from the United States Water Quality Portal [26] (S1 Text). Both external arms are groundwater, so transfer from the fitting archive crosses water-body type as well as geography. Independence rests on archive lineage and a 0.1° cell exclusion between training and external locations, not on a record-level audit, so no fully external claim is made.

The bounded local archives are three openwashdata packages published under CC BY: samples from two 2016 campaigns in Kwale County, Kenya [27]; a laboratory snapshot of 32 Malawian water points, 30 of them boreholes or tubewells, sampled in three rounds with different parameter suites [28]; and a Malawian conductivity series from loggers in monitoring wells [29]. Surface-water samples were excluded from the

Kwale archive, which records no water-point type. Acceptance checked units, identity and plausibility; no charge-balance or conductivity–TDS consistency check was applied, and no records from other Indian or Kenyan custodians, microbial records included, were used. Harmonisation rules, versions and accepted record counts are in S1 Text.

Thresholds and chemical basis

An exceedance rate is a record fraction, the proportion of accepted records above a threshold, reported with its numerator and denominator. Fluoride above 1.5 mg/L and arsenic above 0.010 mg/L follow the WHO guideline values, the arsenic value being provisional [30]. These WHO values are applied to every archive, whereas CGWB judges Indian drinking suitability against the Indian standard IS 10500 [25], whose limits S1 Text compares with ours. Arsenic values labelled mg/L but implausible as such, from four north-eastern states in 2017, were divided by 1,000 (S1 Text). Nitrate was classified above 50 mg/L as NO₃, the WHO guideline value expressed as nitrate ion [30]. The transfer tasks use nitrate above 10 mg/L as N, the value of the United States maximum contaminant level [31]. That is 44.268 mg/L as NO₃, so no nitrate figure is carried between the two analyses. WHO proposes no health-based guideline value for pH or TDS: 6.5–8.5 is the usual optimum pH range for supply, and water becomes increasingly unpalatable at TDS above about 1,000 mg/L [30]; both are used here as operational ranges. Conductivity above 1,500 µS/cm is an unverified operational project convention, not a health guideline or an Indian drinking-water limit; under the factor CGWB recommends, TDS of about $0.65 \times \text{conductivity}$ [4], it corresponds to about 975 mg/L TDS. The 1 NTU turbidity rule is WHO’s value for supporting effective disinfection [30], applied here to untreated water as a project rule. Microbial results use a 100 mL basis, and faecal coliform is never relabelled as *E. coli*.

Binomial uncertainty on record-level exceedance proportions is described by Wilson 95% intervals [32], which do not remove spatial clustering or sampling-design bias, so rates are reported as monitoring-record signals, not as population prevalence or exposure.

Application analyses

For India we formed state or Union Territory $\times$ parameter cells from the accepted CGWB records. A cell was described when it held at least 30 records at 10 or more sites; its median revisit interval, on a site key built from the station or well identifier and the sample date, and its support for leave-one-state-out and temporal analysis were recorded (S3 Table). A cell is decision-relevant when the Wilson 95% interval on its exceedance rate straddles an action level, set at 0.10 as a stated assumption and not a regulatory value, with counts also reported at 0.05 and 0.25. This classification allocates confirmatory effort between states and

parameters and sets no action; CGWB’s own response to an exceedance is at station level, monitoring
 affected areas as trend wells and new hotspots on a 2 km grid [4]. For Kwale and the Malawi laboratory
 snapshot we report record fractions against the same thresholds. For the 67 Kwale sites sampled in both
 campaigns (66 for nitrate and fluoride) we compared threshold states between campaigns using site medians,
 and for the Malawi logger series we counted changes of state between successive daily site medians.

Model channels and validation designs

Model A uses environmental and terrain covariates and the site coordinate, and Model B uses earlier
 measurements at the same site and needs at least two of them. Both use the histogram-based
 gradient-boosting classifier in scikit-learn [33], with configurations fixed in advance and no hyperparameter
 search. Model A’s configuration is in S1 Text; Model B’s was not retained.

Three validation designs were compared on the seven fitting tasks (Fig 3). Each split the fitting archive
 once into training, validation and test partitions keyed on the record (random-row), the station
 (station-grouped) or a contiguous 3° spatial block; Model A was fitted on 24 environmental, terrain and
 coordinate features, calibrated isotonically on the validation partition and scored once on the test partition
 (S1 Text). Block splitting keeps nearby observations together and is the design used here to address
 generalisation to held-out blocks within the sampled geography [8]; no autocorrelation range was estimated
 to set the block size. ROC-AUC (area under the receiver operating characteristic curve), precision–recall
 AUC, sensitivity, specificity, calibration slope and Brier skill were interpreted separately; precision–recall
 AUC is read against prevalence as its no-skill baseline [34], and a 0.5 cut-point was used diagnostically rather
 than as an operational default. Prevalence here is a task’s base rate, the share of its records above threshold,
 not population prevalence. A calibration slope between 0 and 1 means predicted probabilities are too
 extreme, and a slope at or below 0 means they carry no usable probability ordering. The two channels were
 compared separately, in forward-chained yearly folds on records with at least two earlier same-site
 measurements, with DeLong tests on the pooled predictions (S1 Text). No local recalibration was performed,
 because it needs held-out predictions that were not retained.

Fig 3. Three validation designs. How each design splits monitoring records into training and held-out
 sets; positions carry no geographic meaning. Random-row splitting can place records from one station on
 both sides of the split, station-grouped splitting keeps each station whole, and spatial-block splitting assigns
 contiguous 3° blocks.

A mechanistic ablation tested whether aquifer, soil, land-cover and population variables substitute for raw
 coordinates for conductivity and pH; its result, whose intervals overlap, is in S5 Text, and S6 Fig shows one

illustrative surface, read as an ordering and not as a probability. Its stack comprises the national aquifer classification [35], SoilGrids 2.0 soil properties [36], ESA WorldCover 10 m 2021 v200 land cover [37] and WorldPop population density [38].

External transfer assessment

Transfer was assessed per archive $\times$ parameter pair. For each parameter a separate Model A fit was trained on GEMStat records before a cut year, on 33 features that add nearest-station neighbour features and month to the 24 above (S1 Text). These transfer fits are not the seven validation fits and were not evaluated under the three designs, so the optimism measured under the designs is not carried to the transferred scores. The frozen fits were scored on the external archives with no tuning, threshold selection or recalibration after the external data were seen, once every external site sharing a 0.1° cell with a training station had been discarded. Of 24 possible pairs, 13 were evaluated and 11 are reported as untested rather than negative.

The verdict comes from three interval tests rather than from a threshold on discrimination: whether the site-level rank correlation between predicted risk and observed severity has a 95% site-bootstrap interval wholly above 0, whether the rank correlation within 1° blocks has a block-bootstrap interval wholly above 0, and whether lift in the top predicted decile has a site-bootstrap interval wholly above 1. With fewer than two tests evaluable a pair has *insufficient support*, a gap in the evidence that is never merged with a negative result; if every evaluable test passes, *rank transfer is supported*; if none passes, it is *not supported*; otherwise it is *partial or scale-dependent*. Only the outcome of each test was retained, not its statistic or interval, so a verdict carries no stated sampling uncertainty. For every pair we also recorded threshold discrimination and the Spearman rank correlation between predicted risk and observed severity among detected records, each with a site-clustered bootstrap interval, the calibration slope, Brier skill against a reference that knows the external base rate, and confusion counts at a 0.5 cut-point (S1 Text). The retained calibration intercept was estimated jointly with the slope and so is not calibration-in-the-large, and no criterion for adequate calibration was fixed in advance. A supported verdict authorises prioritised confirmatory sampling only, at the level of observations within the archive.

Technology evidence and action rules

The exact-product catalogue comprises 23 named commercial products assessed against 54 criteria, a 1,242-cell grid in which each cell records its value, source, locus and evidence class (S4 Table). Not reported, not researched, conflicting and range-only evidence are kept distinct, and evidence about an adjacent product cannot verify a named model. No record states how the 23 products were chosen or which sources were

searched and when, so the product counts describe the documentary record as located. A criterion meets the documentation threshold when it is documented for at least 12 of the 23 products, counted on its primary database field (strict) or on any database field mapped to it (any admissible). The sufficiency rule requires all 11 active criteria (S1 Text) to be documented; threshold-near classification error, invalid-test rate and testing frequency are not among the active criteria.

Conditional comparison of the ten technology archetypes used the same 11 criteria under seven declared weighting scenarios, with 40,000 Monte Carlo draws per scenario from a Dirichlet distribution centred on the scenario's weights (S1 Text, S7 Table). The technique for order of preference by similarity to ideal solution (TOPSIS) supplied the primary rank [39], and stochastic multicriteria acceptability analysis (SMAA) represented weight uncertainty as a distribution over the feasible weights [40]. The comparison is not screened for the analytes a scenario targets, and no practitioner weights were elicited or invented.

For each monitoring context a rule returns an action (S2 Table): for Kenya and Malawi from the data status alone, whatever the exceedance count, and for India from a per-parameter evidence state that the authors assigned from the transfer verdict, record support and identity limits (S1 Text gives both branches). The evidence states and the rule were recorded on 13 August 2026, after the transfer verdicts had been computed, and no dated protocol records when the action level, support requirement, verdict rule, documentation threshold or active criteria were fixed. The product-level decision is a single determination over the 23-product catalogue, applied to all 26 contexts, not 26 independent determinations.

Reproducibility

Machine-readable result exports were retained for the India and WQP transfers and the mechanistic ablation, and the canonical result tables were verified against them by a script that does not refit; the output file of the validation-design run matches the reported validation results to four decimals. The run scripts are held by the author and were not re-run for this article, no prediction-level file was retained, and no fitted model object is recorded for the validation or transfer models. A result-table verification is not a full training reproduction, and no result in this paper is described as reproduced, replicated or independently re-derived. The scikit-learn version used for the model runs was not recorded (S1 Text).

Artificial-intelligence assistance

We used large language models in preparing this manuscript: OpenAI ChatGPT (version not recorded) to set up the initial project file structure of prompts, rules and scripts, and Anthropic Claude Opus 5, through Claude Code, to write and run data-processing, verification, table- and figure-building scripts (the model

runs were not among them), open cited sources and record the passage supporting each citation, and draft
and revise prose. No AI tool altered or imputed a measurement, and every count, table and quoted value was
checked by script against its source file or recorded passage. S10 Text records each activity, its check and
whether the author has reviewed it; at the time of writing, the author’s review of the prose and of the
citation records was outstanding.

Results

We applied the framework first to India, where the accepted archives are largest, then to the local archives of
Kwale County in Kenya and of Malawi, and finally to published evidence from Bangladesh and Ghana.
Across the 26 monitoring contexts it returned an action and the evidence required to reassess it in every case,
and the 26 took eight distinct actions (Fig 4, Table 1). The most frequent were *local signal plus direct*
confirmation, in 9 contexts, and *monitoring baseline required*, in 5; six of the 26 contexts hold no records, and
their actions follow from that absence. Because the rule returns an action for any data state, this coverage
follows by construction, and the evidence each action rests on is reported below.

Fig 4. Actions returned in the 26 monitoring contexts. Number of contexts, each a country × source
archive × parameter combination, returning each of the eight actions, by country (India 8, Kenya 8, Malawi
10). No context named a product, and the Kenyan and Malawian contexts support no national statement.
Plotted values: S9 Data.

Table 1. Actions and the evidence required to reassess them in the 26 monitoring contexts.
One row per country × source archive × parameter combination; the evidence column names the kind of
evidence that could change the action. The full matrix is S2 Table.

Monitoring context	Records (n)	Action	Evidence required to reassess the action
India, CGWB Year Book: fluoride	35,843	Rank prioritisation plus direct confirmation	Repair longitudinal well identity and run same-sample field-versus-reference validation.
India, CGWB Year Book: nitrate	34,642	Rank prioritisation plus direct confirmation	Repair persistent well/date/sample identity and collect repeat measurements before any temporal model.
India, CGWB portal: arsenic	9,927	Targeted direct reference measurement	Recover method and detection-limit metadata and build a defensible detected/non-detected analytical population.
India, CGWB portal: conductivity	173,029	Rank prioritisation plus direct confirmation	Governance review of the 1,500 µS/cm project convention plus prospective sentinel-versus-spot evaluation; major-ion chemistry at flagged wells.
India, CGWB portal: pH	162,729	Direct measurement required	Collect representative local labels at sentinel sites and assess whether a separate local recalibration or model is useful; pH departures reported by direction.
India, CGWB portal: TDS	30,151	Rank prioritisation plus direct confirmation	Prospective paired conductivity/TDS sampling across relevant water matrices and seasons; recovery of how the portal TDS values were derived.
India, no accepted archive: <i>E. coli</i>	–	Monitoring baseline required	Acquire traceable India microbial measurements with household/source context and compatible denominators.
India, no accepted archive: turbidity	–	Monitoring baseline required	Acquire a machine-readable India archive with site, date, method, unit and valid turbidity results.

Monitoring context	Records (n)	Action	Evidence required to reassess the action
Kenya, Kwale groundwater samples: arsenic	140	Local signal plus direct confirmation	Repeat same-site arsenic measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: conductivity	140	Local signal plus direct confirmation	Repeat same-site conductivity measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: <i>E. coli</i>	–	Monitoring baseline required	Acquire traceable local <i>E. coli</i> records with site, date, unit, method and quality-control fields.
Kenya, Kwale groundwater samples: fluoride	139	Local signal plus direct confirmation	Repeat same-site fluoride measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: nitrate	139	Local signal plus direct confirmation	Repeat same-site nitrate measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: pH	140	Local signal plus direct confirmation	Repeat same-site pH measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: TDS	–	Monitoring baseline required	Acquire traceable local TDS records with site, date, unit, method and quality-control fields.
Kenya, Kwale groundwater samples: turbidity	–	Monitoring baseline required	Acquire traceable local turbidity records with site, date, unit, method and quality-control fields.
Malawi, laboratory snapshot: arsenic	10	Data repair required	Recover arsenic units, method, LOD/LOQ and non-detect semantics before threshold analysis.
Malawi, laboratory snapshot: conductivity	31	Data repair required	Recover the conductivity scale/unit convention and raw instrument export; do not repair by assumption.
Malawi, laboratory snapshot: <i>E. coli</i>	–	Organism-specific baseline required	Acquire organism-specific <i>E. coli</i> measurements with sample volume, reference method and response pathway.
Malawi, laboratory snapshot: faecal coliform	19	Organism-specific baseline required	Acquire organism-specific <i>E. coli</i> measurements with sample volume, reference method and response pathway.
Malawi, laboratory snapshot: fluoride	2	Data repair required	Recover the fluoride missing-value codebook and valid observations before threshold analysis.
Malawi, laboratory snapshot: nitrate	31	Local signal plus direct confirmation	Expand repeat-site nitrate coverage and retain method, calibration, operator and quality-control metadata.
Malawi, laboratory snapshot: pH	31	Local signal plus direct confirmation	Expand repeat-site pH coverage and retain method, calibration, operator and quality-control metadata.
Malawi, laboratory snapshot: TDS	31	Local signal plus direct confirmation	Expand repeat-site TDS coverage and retain method, calibration, operator and quality-control metadata.
Malawi, laboratory snapshot: turbidity	22	Local signal plus direct confirmation	Expand repeat-site turbidity coverage and retain method, calibration, operator and quality-control metadata.
Malawi, logger series: conductivity	714	High-frequency sentinel plus chemical confirmation	Pair the two persistently high-conductivity sites and comparison sites with direct chemistry, calibration, fouling, downtime and energy records.

1. **Every context also abstains at product level:** the framework names no product in any of the 26 contexts, one determination over the 23-product catalogue.
2. **An en dash means the record count is not available:** no accepted measurement population exists in the archives analysed. It does not mean zero exceedances, and absence of records is not evidence of safety.
3. Kwale County and the two Malawi archives are bounded local archives; **none is a national estimate.**
4. Records are accepted records, except for Malawi conductivity, fluoride and arsenic, which failed acceptance and are counted as rows held, and the logger series, counted as site-days. Faecal coliform is its own context and is never relabelled as *E. coli*. Malawi TDS keeps its recorded action although its records share the mixed scales that returned data repair for conductivity, so no TDS count is reported.
5. The evidence column states what would change an action, not the action it would change to, who supplies the evidence, the sample size, frequency, reference method or cost. For Indian conductivity, TDS and pH it adds a clause not in the context matrix (S2 Table), which keeps the recorded text.
6. *Local labels* are direct measurements used to check a model's scores; for Indian conductivity, *governance review* means replacing the project convention with the action value a programme uses; no reference method is named for a *traceable reference workflow*. LOD/LOQ, limits of detection and quantification. The actions are defined in S2 Table.

Record support and decision-relevant cells in the Indian archives

The Indian archives resolve to 186 state or Union Territory × parameter cells in 31 states and Union Territories, of which 160 met the support requirement for description. Accepted record counts range from 173,029 for conductivity to 9,927 for arsenic (Table 2).

Table 2. The Indian evidence base by parameter. One row per parameter, aggregated from 186 state or Union Territory × parameter cells. Exceedance rates are record fractions with Wilson 95% intervals; a decision-relevant cell is one whose interval straddles an assumed 0.10 action level.

Parameter	Accepted records (n)	Records above threshold (n)	Unique sites (n)	Exceedance threshold	Record fraction [95% Wilson interval]	Cells supporting description (of 31)	Decision-relevant cells (n)	Cells supporting leave-one-state-out (of 31)
Conductivity	173,029	39,115	24,128	>1,500 μ S/cm (operational convention)	0.2261 [0.2241, 0.2280]	30	1	22
pH	162,729	10,409	23,704	outside 6.5–8.5 pH units	0.0640 [0.0628, 0.0652]	30	4	21
Fluoride	35,843	2,344	32,561	>1.5 mg/L	0.0654 [0.0629, 0.0680]	27	6	14
Nitrate	34,642	6,403	32,206	>50 mg/L as NO ₃	0.1848 [0.1808, 0.1890]	27	2	18
TDS	30,151	5,442	14,632	>1,000 mg/L	0.1805 [0.1762, 0.1849]	27	4	14
Arsenic	9,927	31	7,931	>0.010 mg/L	0.0031 [0.0022, 0.0044]	19	0	0

1. **Median revisit interval.** Annual or longer in all 145 of 145 cells with a recorded median (conductivity 31; pH 31; fluoride 23; nitrate 23; TDS 30; arsenic 7), and **48 are exactly 365 days**; the 46 from the Year Book tables cannot fall below the edition spacings (365, 731, 1,096 days). The interval describes the archive as held and prescribes no sampling frequency.
2. A cell is **decision-relevant** when its Wilson interval straddles an assumed 0.10 action level: **17 of 160** cells that support description. The intervals treat records as independent, so the count is a lower bound. The classification is not a ranking of severity, prevalence or exposure, and a cell is not a well or a person.
3. Leave-one-state-out support is the record support an evaluation on an unseen state would need; support for description is not support for generalisation, and **for arsenic that count is zero**.
4. The **1,500 μ S/cm** conductivity value is an operational project convention, not a health guideline.
5. Nitrate is classified at 50 mg/L **as NO₃**; the transfer analysis uses 10 mg/L as N, the United States maximum contaminant level, so nitrate figures are not carried between the two.
6. Fluoride and nitrate come from the CGWB Year Book tables and the other four parameters from the CGWB portal. Records come from all 31 states and Union Territories for conductivity, pH, fluoride, nitrate and TDS; arsenic records come from 23 of them. Per-cell counts, intervals and revisit intervals are in S3 Table.

At an assumed action level of 0.10, 17 of the 160 eligible cells are decision-relevant, in 15 states and Union Territories (Fig 5): 6 fluoride cells, 4 pH, 4 TDS, 2 nitrate and 1 conductivity, and none of the 19 eligible arsenic cells. These 17 cells mark where further records could change a state-level classification, not the highest exceedance rates or a ranking of places by hazard, and no action in Table 1 depends on them.

Fig 5. Decision-relevant state or Union Territory × parameter cells in the Indian archives. The 17 cells whose Wilson 95% interval on the exceedance proportion straddles an assumed 0.10 action level, numbered from the widest interval to the narrowest, an order of uncertainty and not of severity. Markers are anchors inside each state or Union Territory, not monitoring sites. Nitrate uses 50 mg/L as NO₃; conductivity uses 1,500 μ S/cm, an operational convention and not a health guideline. Boundaries: Natural Earth (public domain). Plotted values: S9 Data.

Each of the 17 Wilson intervals straddles the action level (Fig 6); the widest, for pH in Nagaland, rests on 44 accepted records and the narrowest, for conductivity in Uttar Pradesh, on 7,280. The count depends on the level: at 0.05 it is 29 cells in 16 states and Union Territories, and at 0.25 it is 4 cells in 4. The intervals treat records as independent although sites are revisited, so the counts are lower bounds, and a cell resolved below the action level is not one where monitoring may stop.

Fig 6. Exceedance proportions for the 17 decision-relevant cells. Exceedance proportion with its Wilson 95% interval for each cell of Fig 5, under the same numbering; n is the number of accepted records, and every interval straddles the 0.10 action level. The intervals treat records as independent although sites are revisited, so 17 is a lower bound. TDS, total dissolved solids. Plotted values: S9 Data.

The other 143 eligible cells are resolved, with intervals wholly above or wholly below the action level. Rajasthan shows the upper side. Of 12,349 conductivity records at 1,641 sites between 2000 and 2024, 6,676 exceeded the 1,500 $\mu\text{S}/\text{cm}$ project convention (54.1%, Wilson 95% interval 53.2% to 54.9%), and of 3,131 TDS records at 1,085 sites, 1,627 exceeded 1,000 mg/L (52.0%, 50.2% to 53.7%). No Rajasthan conductivity or TDS record states its analytical method.

Record support and revisit cadence set what further analysis the archives can carry (Fig 7). Record support would permit a leave-one-state-out evaluation in 89 of the 186 cells, including 9 of the 17 decision-relevant cells, and in none of the 31 arsenic cells. In all 145 cells with a recorded median revisit interval, the median is annual or longer; 48 are exactly 365 days and the longest is 1,534 days. Of the 145, 46 come from Year Book tables, whose intervals cannot fall below the spacing between editions, and some medians rest on a single interval, so the result describes the archive as held rather than the programme as now designed; no action in Table 1 prescribes a sampling frequency. The arsenic record fraction of 0.0031 is not interpretable as a monitoring signal until units, method and detection limits are recovered.

Fig 7. Record support and revisit cadence in the Indian archives. (A) Cells per parameter, of 31, supporting description (grey bar), a leave-one-state-out evaluation (hatched bar) and decision-relevant cells (diamond): 160, 89 and 17 of 186 in all; the numbers beside each bar give the same three counts. (B) Median revisit interval in the 145 cells with one recorded: none is below 365 days and 48 are exactly 365; the 46 Year Book cells cannot fall below the spacing between editions. TDS, total dissolved solids. Full table: S3 Table. Plotted values: S9 Data.

Validation, external transfer and the Indian actions

Validation design changed the performance estimate for the same model specification on the same data, refitted on each design's training partition (Fig 8, Table 3). Across seven tasks from the multinational fitting archive, mean ROC-AUC on the test partition was 0.9396 under random-row splitting, 0.8688 under station-grouped splitting and 0.7696 under 3° block splitting, an optimism of 0.1701 computed from

unrounded means. The figure comes from one partition per design on seven tasks from one predominantly surface-water archive, with no interval. The loss from random-row to block splitting ranges from 0.0768 for fluoride to 0.2927 for nitrate.

Fig 8. Discrimination of the fitting-archive models under three validation designs. Test-partition ROC-AUC (area under the receiver operating characteristic curve) for the seven classification tasks, with means of 0.9396 (random-row), 0.8688 (station-grouped) and 0.7696 (spatial-block). Optimism is the random-row mean minus the spatial-block mean, computed from unrounded values. Values are from retained result tables (Table 3). Plotted values: S9 Data.

Table 3. Model discrimination and calibration under three validation designs, with a paired model comparison. One row per parameter. Prevalence and test-partition ROC-AUC under random-row, station-grouped and spatial-block splitting; PR-AUC and the calibration slope are for the spatial-block design. The last three columns compare Model A and Model B in forward-chained yearly folds.

Parameter	Prevalence	ROC-AUC random row	ROC-AUC station grouped	ROC-AUC spatial block	Optimism	Spatial PR- AUC	Spatial calibra- tion slope	Model A ROC- AUC	Model B ROC- AUC	Model B – A
Arsenic	0.0833	0.9626	0.9230	0.8160	0.1466	0.2710	0.8922	0.9302	0.9530	+0.0228
Nitrate	0.0263	0.9727	0.8527	0.6800	0.2927	0.0743	0.8363	0.9444	0.9779	+0.0335
Conductivity	0.0416	0.9854	0.9411	0.8596	0.1258	0.3428	0.7001	0.9633	0.9858	+0.0225
Suspended solids	0.2277	0.8683	0.8203	0.7463	0.1221	0.4633	0.8541	0.8508	0.8724	+0.0215
Turbidity	0.8323	0.9293	0.8615	0.7440	0.1853	0.9272	0.6749	0.9121	0.9426	+0.0305
Fluoride	0.0228	0.9830	0.9336	0.9062	0.0768	0.1912	0.1925	–	–	–
pH	0.0715	0.8760	0.7496	0.6348	0.2412	0.1855	0.2979	0.8298	0.8940	+0.0643
Mean of 7	n/a	0.9396	0.8688	0.7696	0.1701	n/a	n/a	n/a	n/a	n/a

1. **Optimism is random-row minus spatial-block ROC-AUC**, computed from unrounded values, so it can differ by 0.0001 from a difference of the rounded values shown. Random-row values show, by difference, how much record-level splitting inflates the performance estimate; only spatial-block splitting speaks to held-out areas within the sampled geography, and **none of the designs is a claim about transfer to an unmonitored country**.
2. Model A (33 features, including neighbour features; a separate fit from the validation-design columns, S1 Text) and Model B (12 site-history features) were compared in **forward-chained yearly folds** on records with at least two earlier measurements at the site. Model B exceeded Model A on all six paired parameters, in discrimination and in closeness of the calibration slope to 1 (Model A against Model B: arsenic 0.7608 against 0.8883; nitrate 0.8284 against 0.9344; conductivity 0.7922 against 0.9603; suspended solids 0.8700 against 0.9695; turbidity 0.7689 against 0.9156; pH 0.8442 against 0.9570). The comparison is not evidence of spatial generalisation, and the difference under station-grouped or spatial-block splitting was not estimated. Model B needs earlier measurements at the site, so its advantage is unavailable where monitoring is absent.
3. DeLong p-values are not reported: the exported value, 0.00e+00, is below display precision for tests on 174,533 to 803,735 paired observations, and the test assumes independent observations, which revisited stations violate; at these sizes a test reflects sample size, not practical importance. The ‘Model B – A’ column, +0.0215 to +0.0643, is the quantity to read.
4. **An en dash means not available** (fluoride has no paired comparison); **‘n/a’ means not applicable**. Prevalence is a task’s base rate, the share of its records above threshold, not population prevalence.
5. No mean is given for the spatial calibration slope (0.1925 to 0.8922) or PR-AUC (0.0743 to 0.9272); read PR-AUC against the prevalence in the same row.
6. Values are cited from frozen result tables; **the runs were not re-run for this article** and no prediction-level file was retained. A modelled cell is not a well, tap, source, household or person.

Under block splitting, precision–recall AUC exceeded each task’s overall prevalence for all seven tasks, from 0.0743 for nitrate to 0.9272 for turbidity (Fig 9). Turbidity is the one task in which exceedance is the majority class, with a prevalence of 0.8323. The no-skill baseline is strictly the prevalence of the scored test partition, which the retained tables do not report.

Fig 9. Spatial-block PR-AUC against each task’s prevalence. Precision–recall AUC (PR-AUC) under spatial-block splitting for the seven classification tasks, beside each task’s overall prevalence; the no-skill baseline is strictly the prevalence of the scored test partition, which is not reported. Prevalence is the share of a task’s records above its threshold, not population prevalence. Plotted values: S9 Data.

In forward-chained yearly folds, Model B exceeded Model A on each of the six parameters with a paired comparison, by 0.0215 to 0.0643 in ROC-AUC (Fig 10), an advantage that needs earlier measurements at the same site.

Fig 10. Model A and Model B compared in forward-chained yearly folds. ROC-AUC of Model A (environmental, positional and neighbour features) and Model B (antecedent measurements at the same site) for the six tasks with a paired comparison, each year predicted from earlier years on records with at least two earlier same-site measurements. The comparison is not evidence of spatial generalisation. Plotted values: S9 Data.

Of the 13 archive $\times$ parameter pairs evaluated for transfer, ordering transferred for five (Fig 11): conductivity and TDS in the CGWB portal, fluoride and nitrate in the CGWB Year Book, and TDS in WQP. TDS was the only parameter supported in both archives in which it was tested (threshold ROC-AUC 0.7081 in the CGWB portal and 0.7406 in WQP). Ordering was not supported for pH in either archive in which pH was tested (threshold ROC-AUC 0.4870 in the CGWB portal and 0.4886 in WQP). Arsenic in the CGWB portal had insufficient support, with one evaluable test on 119 scored observations at 116 sites, and its ordering was inverted: ROC-AUC 0.3170 (95% interval 0.2043 to 0.4450). The nitrate ordering was verified at 10 mg/L as N, and its use against 50 mg/L as NO_3 assumes that the ordering does not depend on that difference. Scored observations are fewer than accepted records for every Indian pair; the site exclusion accounts for part of the difference, and the rest is not recorded.

Fig 11. Transfer verdict for each external archive and parameter. Verdict, threshold ROC-AUC and number of external observations for each archive $\times$ parameter pair; 13 of 24 pairs were evaluated, and untested pairs are unknown evidence. The verdict comes from three interval tests, on site-level rank correlation, on rank correlation within 1° spatial blocks and on lift in the top predicted decile, not from ROC-AUC; insufficient support is never merged with not supported. CGWB, Central Ground Water Board; TDS, total dissolved solids. Plotted values: S9 Data.

The rank correlation between predicted risk and observed severity among detected records, which is not one of the verdict tests, had a 95% interval wholly above 0 for 9 of the 13 pairs, including four without supported transfer. Among the five supported pairs it ranged from 0.2025 for nitrate in the CGWB Year Book to 0.4778 for TDS in WQP, and their threshold ROC-AUC ranged from 0.6107 to 0.7426 (Fig 12). For India, the four supported Indian pairs give an ordering that can set the sequence of confirmatory sampling within each archive.

Probability did not transfer with ordering (Fig 13). The calibration slope was below 1 on all 13 evaluated

Fig 12. Threshold discrimination and priority ordering for the 13 evaluated pairs. (A) Threshold ROC-AUC with its 95% site-clustered bootstrap interval (2,000 replicates; 1,998 for Water Quality Portal nitrate); the CGWB portal pH interval, 0.4784 to 0.4964, lies wholly below 0.5. (B) The Spearman rank correlation between predicted risk and observed severity among detected records, with its site-clustered 95% interval; it is not one of the verdict tests. Severity is $\log_{10}$ concentration, or for pH the distance from the centre of the 6.5–8.5 band. Marker shape and colour give the verdict. Plotted values: S9 Data.

pairs, between 0.2383 and 0.4017 on the five pairs whose ordering transferred, and at or below 0 on 4. Within the fitting archive the block-design slopes were already below 1 for all seven tasks, between 0.1925 and 0.8922. Brier skill was negative on all six Indian pairs and on 11 of the 13 overall; the two non-negative values, 0.0470 for *E. coli* and 0.0169 for arsenic in WQP, belong to pairs without supported rank transfer. The conductivity base rate was 0.0638 in the transfer training data and 0.2364 among the scored CGWB portal observations, against 0.2261 among all accepted conductivity records. Calibration-in-the-large was not assessed and local recalibration was not run, so no transferred score can be read as a local exceedance probability.

Fig 13. Calibration slope and Brier skill for the 13 evaluated pairs. (A) Calibration slope, below 1 on all 13 pairs and at or below 0 on 4. (B) Brier skill against a reference that knows the external base rate, negative on 11 of 13 pairs and on all six Indian pairs. Calibration-in-the-large was not assessed and no local recalibration was run, so no pair licenses a model probability. The fitting archive is predominantly surface water and both external archives are groundwater; nitrate is scored at 10 mg/L as N. Plotted values: S9 Data.

At a fixed 0.5 cut-point most exceeding observations were not flagged; for conductivity in the CGWB portal, sensitivity was 0.0787 (S1 Text gives the counts).

For India the rule returned four actions (Table 1). Conductivity, TDS, nitrate and fluoride returned *rank prioritisation plus direct confirmation*: ordering may set the order in which sources are confirmed, but every flagged source still needs a direct measurement, low-ranked sources stay under routine measurement, and no ranking is released with this article. pH returned *direct measurement required*, and arsenic *targeted direct reference measurement*, pending method and detection-limit metadata, with no basis set for choosing targets and laboratory access and cost not assessed. Turbidity and *E. coli*, with no accepted Indian measurement population in the archives analysed, returned *monitoring baseline required*; CGWB lists microbial contaminants among additional parameters to be monitored [4], so the evidence may lie with drinking-water agencies rather than in a new survey. Tables 1 and 4 give the evidence that would change each action.

Repeat measurement in Kwale County, Kenya

The Kwale archive describes groundwater and springs sampled in Kwale County and supports no statement about Kenya as a whole. It records 81 sampling locations (Fig 14) across 71 named localities, 73 of which remain once surface-water samples are excluded, visited in March and June 2016, with 139 or 140 records per

parameter. Locations are counted as distinct recorded coordinates, so the count depends on how coordinates
 were recorded. Measured pH lay outside 6.5–8.5 in 41 of 140 records (29.3%, Wilson 95% interval 22.4% to
 37.3%), all of them below 6.5, and conductivity exceeded the 1,500 $\mu\text{S}/\text{cm}$ convention in 19 of 140 (13.6%,
 8.9% to 20.2%). Nitrate exceeded 50 mg/L as NO_3 in 5 of 139 (3.6%, 1.5% to 8.1%), fluoride exceeded 1.5
 mg/L in 3 of 139 (2.2%, 0.7% to 6.2%) and arsenic exceeded 10 $\mu\text{g}/\text{L}$ in 3 of 140 (2.1%, 0.7% to 6.1%). Most
 locations were sampled twice, so these intervals understate the uncertainty.

Fig 14. Sampling locations of the Kwale County archive, Kenya. The 81 sampling locations of the Kwale County archive, from two 2016 campaigns [27], reprojected from UTM zone 37S on the Arc 1960 datum without a datum shift; no coastline is drawn at this scale. Surface-water locations are drawn; the counts in the text exclude them. Inset: Kenya, with the sampled area outlined. The archive is local and supports no national statement; the openwashdata package is published under CC BY 4.0. Plotted values: S9 Data.

Two of the three fluoride exceedances, and two of the 19 conductivity exceedances, come from one
 thermal spring with a recorded temperature of about 59 °C, and the third fluoride exceedance and one June
 conductivity crossing are unverified values (S1 Text). The counts include these records as the archive holds
 them.

Of the 67 sites sampled in both campaigns (66 with nitrate and fluoride results in both), 15 of the 24 with
 pH outside the range in March were still outside it in June and 2 more crossed; 8 of the 10 above the
 conductivity convention in March remained above and 1 more crossed; and for nitrate, fluoride and arsenic
 one site was above the threshold in both campaigns, with 2, 1 and 1 crossing only in June (S11 Table); the
 fifth nitrate exceedance is at a location sampled once. The source labels the campaigns dry-season and
 wet-season, but one campaign in each season of a single year does not describe a seasonal cycle.

No model was used for Kwale. For pH, conductivity, nitrate, fluoride and arsenic, the rule returned *local
 signal plus direct confirmation* from the data status alone: repeat measurement at the same sites across
 further seasons, with threshold classifications confirmed by a reference method the rule does not name
 (Table 1). Contexts that share the archive and the action could be met by one sampling round. The action
 sets what to measure next; it does not postpone confirmation at, or a programme’s response to, a source
 recorded in 2016 above a WHO health-based guideline value. TDS, turbidity and *E. coli*, for which the
 archive holds no records, returned *monitoring baseline required*.

Data repair and sentinel monitoring in Malawi

The Malawi laboratory snapshot holds one record per water point for 32 water points sampled between
 December 2017 and July 2019 (Fig 15), and it is not a national sample. Measured pH lay outside 6.5–8.5 in 7

of 31 records (22.6%, Wilson 95% interval 11.4% to 39.8%), all between 6.24 and 6.42, and the archive flags each as within the Malawian standard it applies. No nitrate record of 31 exceeded 50 mg/L as NO₃, which leaves an upper Wilson bound of 11.0%. Turbidity exceeded the 1 NTU rule in 2 of 22 records (9.1%, 2.5% to 27.8%), both flagged as within the Malawian standard. Faecal coliform was detectable in 4 of 19 samples, all collected in February 2019 and none with a recorded analysis date. Faecal coliform is not *E. coli*, but WHO treats thermotolerant coliforms as an acceptable alternative indicator of faecal pollution [30], so detection at those four water points would ordinarily warrant inspection and resampling whatever the organism-specific evidence gap.

Fig 15. Malawian laboratory water points and conductivity logger sites. The laboratory snapshot of 32 Malawian water points [28] and the conductivity logger series at 19 sites [29]. Nearby markers overlap at this scale, and neither archive is a national sample. Boundaries: Natural Earth (public domain); the openwashdata packages are published under CC BY 4.0. Plotted values: S9 Data.

Four parameters could not be read as recorded. Conductivity failed a unit-consistency check, TDS is recorded on the same mixed scales in 10 of 31 records, so no TDS count is reported, fluoride held only a placeholder code, and arsenic had no documented unit. Conductivity, fluoride and arsenic returned *data repair required*, to be made from recovered records and not by assumption; only the originating laboratory or the data publisher can make it, and the rule has no fallback if the records cannot be recovered. The laboratory pH, TDS, nitrate and turbidity contexts returned *local signal plus direct confirmation*, although under the conductivity rule the TDS context would return data repair, and for nitrate the action amounts to wider coverage. *E. coli*, with no organism-specific measurement, and the faecal coliform context both returned *organism-specific baseline required* (Table 1).

The logger extract holds conductivity from 19 sites in 9 districts, reduced to 714 daily site medians between 31 December 2024 and 7 February 2025, a window of 39 days within Malawi’s wet season from November to April [41]. On 76 of the 714 site-days, conductivity exceeded the 1,500 µS/cm convention: two sites, both in Chikwawa District, exceeded it on every recorded day and no other site on any day, so across 695 successive site-day transitions no site changed state. The two sites have median conductivities of about 5,690 and 2,160 µS/cm, and the next highest site median is about 1,360 µS/cm. The logger does not identify the cause, since conductivity is not specific to any contaminant, does not record whether values are temperature-compensated, and says nothing about the dry season. The rule returned *high-frequency sentinel plus chemical confirmation*, pairing the two persistent sites, and comparison sites, with direct chemistry and with instrument calibration, fouling, downtime and energy records; the analytes, comparison sites, chemistry frequency and logging period are not set by this study.

Screening and service design from published evidence in Bangladesh and Ghana

In Bangladesh and Ghana no archive was analysed and no model was transferred, and neither setting is among the 26 contexts. The Bangladesh decision is whether a field kit can classify a well near the arsenic threshold accurately enough to leave it in service. In a blind evaluation of 123 wells against laboratory measurement, with 65 tested twice, one kit classified 110 wells correctly against the WHO guideline of 10 µg/L and 113 against the Bangladesh standard of 50 µg/L [42]. All 13 of its misclassifications against 10 µg/L were underestimates, in wells between 10 and 27 µg/L [42]. Against 50 µg/L its over- and underestimates were evenly distributed, and the evaluation omitted the kit's hydrogen-sulphide oxidation step [42]. A comparison of eight commercial kits with hydride-generation atomic absorption spectrometry found two accurate and precise in tests that controlled for colour-matching error, four accurate or precise but not both, and two neither [43].

Applied to that evidence, the framework permits screening under a stated decision rule with confirmation of screen positives and of a sample of negatives by a reference method. The threshold and the fraction of negatives confirmed are not set by this study; at 10 µg/L the recorded errors all missed contaminated wells, which is the case for confirming negatives, whereas at 50 µg/L they ran in both directions. Kit and laboratory costs in Bangladesh were not assessed. The evidence that would change the action is a false-negative rate near the threshold, with its interval, under the rule a programme uses, together with operator and invalid-test rates.

The Ghana decision is how to organise *E. coli* testing so that frequency, transport, incubation and staffing remain affordable. Modelled costs per membrane-filtration test for four service approaches, in three study areas in Ghana and Uganda, favoured centralised testing in two of the three [44]. In the Ghanaian study area, centralised testing was cheapest in 96% of simulations, at a median of USD 10 per test [44]. The same study reports the WHO recommendation that samples not be stored for more than 6 h before processing, which links distance to the choice of service, and publishes its cost model for others to run [44]. A review of commercial field tests found none that met the UNICEF target product profile, with the need for a power source the largest limitation and 18 to 24 h of incubation still required [45].

The framework therefore treats the instrument as one component of a service architecture chosen on cost per valid result, which counts invalid and repeat tests; a valid microbial result is one processed within the holding time, incubated as specified and accompanied by passing controls. Until that is available, modelled cost per test by service approach, bounded to its study areas and assumptions, is the basis a programme has. Neither costing study states a price year, although one converted costs collected in December 2014 and January 2015 at 1 January 2015 exchange rates [14], and none of the 23 products in the documentary

catalogue documents a per-test cost with its basis, currency and date. Neither setting’s evidence supports
naming a kit or a product.

Product documentation and actions across settings

Across the settings, the technology question meets the same documentary record. The exact-product grid holds 23 commercial products against 54 criteria, 1,242 cells (Fig 16). Of these, 891 (71.7%) are not reported in any located document, 195 (15.7%) are document-sourced, 9 of them from regulatory certification, and 106 were not researched, all for two products. Not reported means that the located documentation is silent: the criterion may still have been measured, and no product has been found wanting. The 9 regulatory cells all belong to one product; they are independent of the manufacturer but are documentation, not device-versus-reference testing. Among the records assembled for the 23 products, paired device-versus-reference field evidence was located only for arsenic, and only one of the three recorded comparisons has recoverable publication provenance.

Fig 16. Documentary evidence for 23 named products across 54 criteria. The 1,242 cells of 23 products × 54 criteria: 891 not reported in any located document, 106 not researched (all for two products), 40 conflicting, adjacent-product or range-only, 10 not applicable, and 195 document-sourced, 9 of them from regulatory certification, not device-versus-reference testing. A not-reported cell is unknown evidence, not evidence of poor performance. Product identities and sources: S4 Table. MTBF, mean time between failures; QC, quality control; CV, coefficient of variation; IP, ingress protection. Plotted values: S9 Data.

Under the two counting rules set out in Materials and methods, 1 of the 11 active criteria meets the documentation threshold on the strict field and 4 on any admissible field (Fig 17). No product documents all 11 active criteria, including the 21 that were researched. On any admissible field the most any product documents is 9, reached by 1 product, with 4 products documenting 8 and 4 documenting 7; on the strict field the most is 7. The outcome therefore rests on requiring all 11: a rule requiring 9 would admit one product. For 19 of the 54 criteria, no located document covers any product. In all 26 contexts, the framework named no product: one determination over the 23-product catalogue applied to every context, not 26 separate determinations and not evidence about the contexts.

Fig 17. The 11 active criteria against the documentation threshold. Products documenting each active criterion on strict primary fields and on any admissible field, against a threshold of 12 of 23 products: 1 criterion clears it on strict fields and 4 on any admissible field. No product documents all 11 and the most any documents is 9, so the framework returns a product-level abstention, a statement about the evidence base and not a ranking. Plotted values: S9 Data.

Among technology archetypes, a population separate from the named products, a single archetype led under every cost mapping in 5 of 7 weighting scenarios, and the leader changed with the cost mapping in the

other two (S7 Table); because the comparison was not screened for analytes, a leading archetype is preference context under stated weights, not a product or the class of instrument to use. Table 4 brings the settings together: for each, the evidence held, what a model may and may not do, the action returned and the evidence that would change it; S12 Table adds the monitoring decision each setting poses and where the framework declines.

Table 4. Evidence, model role, action and reassessment evidence in each application setting. One row per setting. Bangladesh and Ghana enter through published evidence only. Table 1 lists the 26 monitoring contexts behind the India, Kenya and Malawi rows.

Setting	Evidence held	What a model may do	Action returned	Evidence that would change it
India (CGWB portal and CGWB Year Book)	186 cells in 31 states and Union Territories; 160 eligible; 17 decision-relevant at a 0.10 action level, which locate where further records could change a classification and set no action; median revisit interval annual or longer in all 145 cells with one recorded	Ordering may sequence confirmatory sampling for conductivity and TDS (CGWB portal) and fluoride and nitrate (CGWB Year Book); pH not supported; arsenic insufficient support; no pair licenses a probability	Rank prioritisation plus direct confirmation (4); Monitoring baseline required (2); Direct measurement required (1); Targeted direct reference measurement (1)	Local direct measurements permitting recalibration; persistent well, date and sample identity; arsenic method and detection-limit metadata; accepted turbidity and <i>E. coli</i> populations
Kenya, Kwale County (groundwater and springs)	Records above threshold: pH 41 of 140, conductivity 19 of 140, nitrate 5 of 139, fluoride 3 of 139, arsenic 3 of 140; 67 sites sampled in both 2016 campaigns	None; no model was fitted or scored	Local signal plus direct confirmation (5); Monitoring baseline required (3)	Repeat same-site measurement across further seasons of the year; 9 of 24 sites outside the pH range in March were inside it in June
Malawi, laboratory snapshot	Records above threshold: pH 7 of 31, nitrate 0 of 31, turbidity 2 of 22; faecal coliform detectable in 4 of 19; conductivity, fluoride and arsenic not readable as recorded, and TDS on the same mixed scale as conductivity in 10 of 31 records, so no TDS count is given	None; no model was fitted or scored	Local signal plus direct confirmation (4); Data repair required (3); Organism-specific baseline required (2)	Recovered conductivity and TDS scale, fluoride codebook and arsenic units; organism-specific <i>E. coli</i> measurement; wider repeat-site coverage
Malawi, conductivity logger series	19 sites, 714 site-days over 39 days; 76 site-days above 1,500 $\mu\text{S}/\text{cm}$; 2 sites above on every recorded day; 0 changes of state in 695 transitions	None; temporal description only	High-frequency sentinel plus chemical confirmation (1)	Direct chemistry with calibration, fouling, downtime and energy records at the persistent and comparison sites
Bangladesh (published evidence only)	One kit classified 110 of 123 wells correctly against 10 $\mu\text{g}/\text{L}$ and 113 against the Bangladesh standard of 50 $\mu\text{g}/\text{L}$; all 13 errors at 10 $\mu\text{g}/\text{L}$ were underestimates, and at 50 $\mu\text{g}/\text{L}$ errors ran in both directions; in tests controlling colour-matching error, two of eight kits were accurate and precise	None; no model transferred	Screening under a stated rule, with reference confirmation of positives and of a sample of negatives; the threshold and the fraction confirmed are the programme's to set	A false-negative rate near the threshold, with its interval, under the programme's rule; operator and invalid-test rates

Setting	Evidence held	What a model may do	Action returned	Evidence that would change it
Ghana (published evidence only)	Modelled cost per test favoured centralised testing in two of three study areas in Ghana and Uganda; in the Ghanaian area it was cheapest in 96% of simulations; no reviewed field test met the UNICEF target product profile	None; no model transferred	Choose a service architecture on cost per valid result, with the instrument as one component	Network-specific cost per valid result, including invalid and repeat tests, and the sampling frequency at which onsite testing becomes cheaper

1. Counts above a threshold are record fractions within each archive, not prevalence, and a state or Union Territory cell is not a well or a person. Thresholds are those of Materials and methods, applied to every archive; arsenic 0.010 mg/L is 10 µg/L.
2. Kwale County and the two Malawi archives are bounded local applications and not national samples; the Kwale counts exclude surface water, and two of its three fluoride exceedances come from one thermal spring.
3. Bangladesh and Ghana had no archive analysed and no model transferred; the kit counts are for one kit at the two thresholds stated, and the costs are modelled.
4. 1,500 µS/cm is an operational project convention and not a health guideline. *Ordering* is a model's ranking of sources; ordering that transfers licenses the sequencing of confirmatory sampling, never a probability.
5. No product is named in any setting. S12 Table adds, for each setting, the monitoring decision it poses and where the framework declines.

Discussion

Implications for monitoring practice

Taken together, the applications show which kind of evidence carried each action. The 18 Kenyan and Malawian actions came from record harmonisation and data status alone, and in Malawi harmonisation turned unreadable parameters into a defined data-repair task rather than a silent gap. The Indian actions drew on record support and on the transfer verdicts: external transfer separated an ordering that can sequence confirmatory sampling, in 5 of 13 pairs, from probabilities that no pair supported. The validation designs entered no action. With one partition per design on seven tasks, they showed the performance estimate for the same model specification falling by 0.1701 in mean ROC-AUC from random-row to block splitting, which is why the framework ties a model's permitted use to the design rather than to the metric. The documentary audit turned missing product information into a criterion-by-criterion account of what no located document reports (S4 Table). No action rests on a single metric.

In India the results support a sequence rather than a map. Where it transferred, ordering could decide which conductivity, TDS, nitrate or fluoride sources to confirm first, though not which may be left untested; at the assumed 0.10 action level, the 17 decision-relevant cells, a lower bound, locate where further records could change a state-level classification; and resolved cells such as Rajasthan call for contaminant-specific confirmation rather than more records. Because calibration slopes were below 1 on every transfer pair and no pair was shown to be adequately calibrated, a transferred score should not be read as the probability that a

source exceeds its guideline, and because the transfer also crossed water-body type, the supported pairs do not separate a change of geography from a change of water type. These are implications of retrospective results, not a tested operational gain.

In Kwale County and Malawi no model output entered any decision, and the constraint was the record itself. In Kwale the binding step is returning to the same sources, since pH classifications changed between campaigns at 11 of the 67 sites sampled twice, and the fluoride signal rests on a thermal spring and an unverified value that a practitioner would check before acting. In the Malawi laboratory snapshot, four parameters could not be read as recorded, and three of them returned data repair, which needs units, a missing-value codebook and, for arsenic, detection limits to be recovered; these gaps describe the records and say nothing about the water. The logger series shows persistence at two sites without identifying its cause. Both sites are in Chikwawa District, and groundwater salinity too high for potable use has been documented in Malawi's lower Shire Valley [41]; whether that explains these sites was not assessed. The snapshot's gaps are in line with reports that groundwater monitoring data in Malawi are persistently scarce, a scarcity made worse by existing monitoring networks that are not operating [46].

In Bangladesh and Ghana the constraint lay in the decision rule and the service design rather than in any model. In hypothetical well-switching scenarios for Araihaazar, Bangladesh, relying on kit rather than laboratory data lost only a modest part of the reduction in average arsenic exposure, set against the logistical and financial challenge of supplying laboratory data alone [47]. The same analysis found the 50 $\mu\text{g/L}$ threshold used in Bangladesh, rather than the WHO guideline of 10 $\mu\text{g/L}$, close to optimal for average exposure reduction [47]. A programme could accept kit screening on that basis, but the residual error it will tolerate is the programme's choice and not a result of this study. For Ghana, both costing studies cited here, one across sub-Saharan Africa and one covering study areas in Ghana and Uganda, express cost per test with quality-control samples added as a markup [14, 44], whereas a service design needs cost per valid result, counting invalid and repeat tests.

Relation to validation, calibration and product-evaluation research

Mean discrimination fell from 0.9396 to 0.7696 between two validation designs applied to the same models and data, whereas a review of machine-learning models for United States drinking water found tenfold cross-validation or a randomly chosen test set the most common basis for reported performance [18]. In a forest biomass mapping study, non-spatial validation suggested that a model predicted more than half of the variation, whereas spatial validation found almost no predictive power [48]. A methodological source on spatial dependence records the counter-argument that spatial cross-validation is not the right way to evaluate

map accuracy [9], so the single spatial-block split is used here to address a generalisation claim, not to certify
accuracy away from sampled locations. Whether the national hazard models for India cited in the
Introduction carry such optimism was not tested here. What the framework contributes here is a rule that
ties the validation design to the use a model is allowed, so that a random-row figure cannot authorise
decisions in unsampled places.

The calibration result also has precedents. Estimated risks can be unreliable even when discrimination is
good, for example overestimating risk when a model is used in a setting where incidence is lower [11].
Calibration-in-the-large, the calibration slope and discrimination have been proposed as distinct measures of
model performance, with problems of calibration-in-the-large often found at external validation [49], and
moderate calibration has been proved to guarantee decision-making that does no harm [50]. No transferred
score in this study was shown to be moderately calibrated, so none carries that guarantee, and because
calibration-in-the-large was not assessed, the study shows that transferred probabilities were miscalibrated
without describing every form of that miscalibration. The permission gate turns this distinction into a
decision rule: a model whose ordering transfers may sequence confirmatory sampling, and only local
calibration can license a probability.

Evidence from outside the product catalogue points the same way as the documentary shortfall. Besides
the operator-dependent kit verification cited in the Introduction, a laboratory evaluation of a marketed
low-cost sensor found no correlation between its score and the presence of *E. coli* [51]. Both evaluated one
product at a stated date, and no conclusion about any product is drawn from them. Most of 72 institutions
assessed across ten sub-Saharan African countries did not achieve the microbial testing levels set by
applicable standards or WHO guidelines [52], and drinking-water surveillance often falls short of its required
frequency [2]. Threshold-near performance, operator dependence and achievable testing frequency are what
product documentation would need to report for a programme to rely on a named instrument.

Abstention here differs from its usual machine-learning form, in which a model abstains where it is likely
to make a mistake [17]. The framework abstains where the evidence needed for a decision is absent, so its
abstention describes the evidence base rather than predictive error. Stochastic multicriteria acceptability
analysis aids decisions in problems with inaccurate, uncertain or missing information [40], and the archetype
comparison draws on it, but no sampling of weights can supply a criterion value that no located document
holds. We therefore read the documentary audit as the substantive result, and the product-level abstention
as the output of the sufficiency rule applied to it: it shows which documentation is missing before named
products can be compared.

Extension to new monitoring settings

Using the framework in another monitoring programme would need at least six steps, several of which go beyond what was done in the applications reported here. First, identify and quality-check the local archives and their provenance. Second, align units, analytes and source types, and justify the thresholds locally; this study applied WHO values and, for conductivity, a project convention. Third, assess site identity, record support and spatial and temporal dependence before choosing the validation designs; no autocorrelation range was estimated here. Fourth, validate any model's ordering in the new setting and calibrate it on local measurements before its probabilities are used; no local calibration was run here. Fifth, assemble technology and deployment evidence, including threshold-near performance, invalid-test rates and service constraints, which were not among the active criteria here. Sixth, assess the feasibility of the returned actions with practitioners and test prospectively whether they change monitoring decisions for the better.

Two observations would test the framework itself. It would gain support if the same rules, applied to a catalogue with fuller documentation, admitted a named product, since that would show that the abstention reported here reflects this evidence base rather than rules no product could meet. It would be shown to be underspecified if two analysts applying the stated rules to the same archive returned different actions. No one outside this project has yet applied the rules, the framework has not been compared with an alternative procedure, and no monitoring programme has reviewed the returned actions for feasibility or cost.

The evidence that would change the actions reported here is specific: local measurements that allow model recalibration (S8 Table, priority D4), persistent site and well identity (D2), paired chemistry at sentinel sites (D3), threshold-near kit performance under a stated rule (A1), and invalid and repeat tests recorded in a deployed network, from which cost per valid result follows (C1). The measurements a recalibration needs are themselves direct measurements, so its value to a programme depends on the model then reducing later measurement, which was not assessed. S8 Table sets out 16 research priorities in four families; it does not cover a comparison of service architectures on cost per valid result, the local conductivity–TDS relation, or the two-analyst test proposed above.

Conclusion

The evidence-aware decision framework developed here connects archived monitoring records, model validation and transfer, and instrument documentation to specific monitoring actions, and it states for each action the evidence required to reassess it. Applied retrospectively to 26 monitoring contexts in India, Kwale County in Kenya and Malawi, it returned eight distinct actions, and its decision logic was applied to

published evidence from Bangladesh and Ghana. Across seven fitting-archive tasks, mean ROC-AUC fell from 0.9396 under random-row to 0.7696 under spatial-block splitting, which is why the framework ties a model's permitted use to the validation design rather than to a headline metric. In India, ordering transferred for four of six archive $\times$ parameter pairs and can sequence confirmatory sampling within those archives, and at an assumed 0.10 action level further records could change the classification of at least 17 of 160 eligible cells; no transferred score was used as a probability. In the bounded local archives of Kwale and Malawi, the actions were repeat measurement, baselines, data repair and paired chemistry, and the published evidence from Bangladesh and Ghana supported screening with confirmation and a service design chosen on cost per valid result, a quantity neither costing study reports.

The documentary audit showed which information is missing before named products can be compared: most product documentation cells are not reported in any located document, and no product documents all 11 active criteria. The next step is to apply the framework prospectively with a monitoring programme, collecting the local measurements that allow model recalibration, persistent site identity, paired chemistry, threshold-near kit performance and cost per valid result, and testing whether the returned actions improve confirmatory sampling in practice; the results reported here are retrospective, and no operational effect has yet been tested.

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Supporting information

S1 Text. Harmonisation, model configuration and decision rules. Harmonisation, unit and chemical-basis rules, with the Indian standard's limits and the arsenic unit remedies; the accepted Indian populations and their record counts; the data states recorded in place of removed records; how the 26 monitoring contexts were generated; the versions of the WQP extract and the Malawi snapshot; the validation-design and transfer-fit configurations, interval replicates and confusion counts at the 0.5 cut-point; the decision-rule branches, the 11 active criteria and the dated records of when rules were set; the Kwale values that are not verified; the training-set sizes of the transfer tasks; the Year Book source documents, with their URLs, Internet Archive snapshot identifiers and SHA-256 digests; and the 62 National Water Data Portal resource files, with their byte counts.

S2 Table. The 26 monitoring contexts in full. Every field of the context matrix, of which Table 1 carries the decision columns, and the definitions of the eight actions: one row per country $\times$ source archive $\times$ parameter combination. The product-level decision is one determination applied to every row.

S3 Table. Support, cadence and exceedance for 186 Indian state or Union Territory $\times$ parameter $\times$ source cells. Table 2, Fig 5, Fig 6 and Fig 7 summarise it. Exceedance proportions are of accepted records, with Wilson 95% intervals; a proportion of records is not a proportion of people.

S4 Table. The 23×54 product documentation grid. Product identities (P01–P23 in Fig 16) and, for each cell, the evidence class, value, unit, source document and locus. A not-reported cell is unknown evidence about a product, not evidence that the product performs poorly.

S5 Text. Mechanistic ablation. Rung definitions, configuration, layer sources, and the discrimination of each rung with its block-bootstrap interval.

S6 Fig. An illustrative conductivity prediction surface. A 60 km window centred on 24.0°N, 87.5°E, chosen by the number of monitoring sites with complete coverage of the 16 mechanistic features, without reference to any conductivity value or model output. (A) The output of the coordinate-free full-stack rung (A5) refitted under the frozen configuration and without tuning, with every three-degree block intersecting the window excluded from training — two blocks, because the window straddles 24°N — so no training observation lies inside the area shown: 148,029 training rows, 7,421 test observations at 1,613 test sites, the 1,500 $\mu\text{S}/\text{cm}$ conductivity convention (an operational screening value, not a health guideline) and a training prevalence of 0.2292. The display refit carries no calibration result, so its scale is read as an ordering, not a probability. (B) Soil pH (water), 0–5 cm, from SoilGrids 2.0. (C) Population density in 2020, people per km^2 on a logarithmic scale, from WorldPop. Grey cells hold no value. The land-cover, aquifer, clay and organic-carbon layers of the stack are not drawn, and no monitoring site is shown. The surface is a spatial illustration, and a modelled cell is not a well, tap, household or person.

S7 Table. Rank acceptability of technology archetypes in seven weighting scenarios. For each scenario, the archetype that leads among the ten compared, its mean probability of ranking first and the range of that probability across 13 cost mappings, its expected rank, and whether the same archetype leads under every mapping. The archetypes are a separate population from the named products; no row is a product recommendation, and the product-level outcome is abstention in every context.

S8 Table. Sixteen research priorities in four families. Each priority with its family, priority level, proposed study design, measurement endpoint and the decision it would improve.

S9 Data. The values plotted in the main-text figures and S6 Fig. One CSV file per figure, named by figure and written by the figure builders from the numbers they drew.

S10 Text. Record of artificial-intelligence assistance. Each activity, its inputs and outputs, and the verification applied to it.

S11 Table. Record fractions for the Kenyan and Malawian archives, Kwale paired campaigns and the Malawi logger summary. Eleven record fractions with Wilson 95% intervals, which treat records as independent and so understate uncertainty; threshold states at the Kwale sites sampled in both 2016 campaigns, with surface-water samples excluded; and the summary of the 39-day logger extract, all recomputed from the raw openwashdata archives. Record fractions are not prevalence, and neither country’s archives are national samples. The Malawi TDS fraction is not interpretable, because TDS is recorded on mixed scales in 10 of 31 records, and the article reports no TDS count.

S12 Table. The setting synthesis in full. For each application setting, the monitoring decision it poses, the evidence held, what a model may do, the action returned, the evidence that would change it and where the framework declines. Table 4 carries five of these seven columns, and its notes apply here.